Lecture Notes: Regularization Part 1: Ridge (L2) Regression

I. The Problem: Overfitting and High Variance

This StatQuest assumes you understand **bias vs. variance** and **linear models**.

1. The Ideal Case (Lots of Data)

- If we have a large dataset (e.g., mouse weight vs. size) that looks linear, we can use **Linear Regression (Least Squares)** to fit a line [01:29].
- We find the line that results in the **minimum sum of squared residuals** [01:45].
- The resulting equation, $Size = Intercept + Slope \times Weight$, is a good model. With lots of data, we are confident it reflects the true relationship [02:26].

2. The Problem Case (Small Sample Size)

- What if we only have **two data points** (our "training data")? [02:34]
- Least squares will fit a line *perfectly* through these two points, resulting in a **sum of squared residuals = 0** [02:44].
- However, this new line is very different from the "true" original line [02:54].
- When we test this new line on the rest of the data (the "testing data"), the sum of squared residuals is very large [03:18].
- This is called **overfitting**. The model has **low bias** (it fits the training data perfectly) but **high variance** (it fails badly on new data) [03:27].

II. The Solution: Ridge Regression

1. The Main Idea

- The core idea of Ridge Regression is to find a new line that **doesn't fit the training data as well** [03:41].
- We intentionally introduce a **small amount of bias** into the model [03:49].
- In return for this small bias, we get a **significant drop in variance** [03:58].
- This trade-off (a slightly worse fit on training data) provides much **better long-term predictions** on new, unseen data [04:05].

2. How Ridge Regression Works (The Math)

Ridge Regression changes the "cost function" that we are trying to minimize.

- **Least Squares** minimizes:

  $$SS_{Residuals}$$

  (Sum of Squared Residuals) [04:23]
- **Ridge Regression** minimizes:

  $$SS_{Residuals} + \lambda \times (\text{Slope})^2$$

  [04:44]

This new part is the **Ridge Regression Penalty**.

- λ (lambda) is a value that determines the *severity* of the penalty [05:02].
- The penalty term $(\text{Slope})^2$ "punishes" the model for having a steep slope.

3. A Worked Example

Let's compare the two models on our 2-point training set [05:19]:
(Assume $\lambda = 1$ for now)

A. Least Squares Line:

- $SS_{Residuals} = 0$ (it's a perfect fit)
- Slope = 1.3
- **Ridge Cost Function:** $0 + 1 \times (1.3)^2 = \mathbf{1.69}$ [05:46]

B. Ridge Regression Line:

- This line is "flatter" and doesn't hit the points perfectly.
- $SS_{Residuals} = (0.3)^2 + (0.1)^2 = 0.1$ [05:56]
- Slope = 0.8
- **Ridge Cost Function:** $0.1 + 1 \times (0.8)^2 = 0.1 + 0.64 = \mathbf{0.74}$ [06:25]

Conclusion:
The total cost (0.74) for the Ridge Regression line is *lower* than the cost (1.69) for the least-squares line. Therefore, Ridge Regression **prefers the flatter line**, even though it has a slightly worse residual error [06:56].

III. The Effect of the Penalty

1. Slopes and Sensitivity

- A **steep slope** (like in our overfit line) means the prediction is **very sensitive** to small changes in the input (Weight) [07:54]. This is a sign of high variance.
- A **small slope** (like in the Ridge line) means the prediction is **less sensitive** to changes in the input [08:21].
- The Ridge penalty $\lambda \times (\text{Slope})^2$ forces the model to have a smaller slope, making the predictions **less sensitive** and **reducing variance** [08:34].

2. The Role of Lambda (λ)

λ is the "tuning parameter" that controls this bias-variance trade-off [08:51].

- **If $\lambda = 0$:**
  The penalty term is zero. Ridge Regression is minimizing *only* the $SS_{Residuals}$. The result is **identical to the standard Least Squares line** [09:02].
- **As λ increases:**
  The penalty for having a large slope becomes more severe. To minimize the total cost, the model is forced to make the slope smaller and smaller [09:26].
  - $\lambda = 1 \rightarrow$ Slope is smaller.
  - $\lambda = 2 \rightarrow$ Slope is even smaller [09:41].
  - As $\lambda \rightarrow \infty$, the slope gets asymptotically close to **0** (a flat horizontal line) [09:47].
- **How do we find the best λ?:**
  We don't guess. We **use cross-validation** (e.g., 10-fold cross-validation) [10:09]. We try a bunch of different values for λ and choose the one that results in the lowest variance (i.e., the best performance on the hold-out folds) [10:02].

IV. Generalizing Ridge Regression

1. Other Model Types

Ridge Regression is not just for simple $y = mx + b$ lines. It can be applied to many models:

- **Discrete Variables:** If predicting `Size` from `Diet` (Normal vs. High-Fat), the model is $Size = \text{Avg}_{\text{Normal}} + \text{DietDiff} \times \text{HighFat}$. Ridge Regression will shrink the `DietDiff` parameter towards zero [11:12].
- **Logistic Regression:** When predicting a probability (e.g., *obese* or *not obese*), Ridge Regression shrinks the slope of the logistic curve [13:24]. (Note: It optimizes the *sum of likelihoods* instead of squared residuals, since logistic regression uses Maximum Likelihood)

[13:49].

2. The General Penalty (Multiple Regression)

What if we have a very complex model with many parameters?

$$Size = \beta_0 + \beta_1 \times \text{Weight} + \beta_2 \times \text{Diet} + \beta_3 \times \text{Age} + \ldots$$

- β_0 is the y-intercept.
- $\beta_1, \beta_2, \beta_3 \ldots$ are the slopes (parameters) for each variable.

The **general Ridge Regression penalty** contains *all* parameters *except* the y-intercept:

$$\text{Ridge Cost} = SS_{Residuals} + \lambda \times (\beta_1^2 + \beta_2^2 + \beta_3^2 + \ldots)$$

[14:38]

- This is also known as the **L2 Norm**, which is why Ridge Regression is often called **L2 Regularization**.
- **Why is the intercept (β_0) not included?** The intercept is just a baseline; it's not "scaled" by any measurement [15:08]. Penalizing it would not make sense. We only want to penalize the "sensitivity" of our predictions to the input variables.

V. The "Magic" of Ridge Regression: Solving the Unsolvable

This is the coolest property of Ridge Regression.

- To solve for parameters in **Least Squares**, you **must have more samples (data points) than parameters (features)**.
  - To find 2 parameters (slope, intercept), you need at least 2 data points [15:35].
  - To find 3 parameters (e.g., $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$), you need at least 3 data points [17:08].
  - To find 10,001 parameters (e.g., predicting `Size` from 10,000 genes), you need at least 10,001 data points (mice) [17:24].
- **The Problem:** In modern fields like genetics, it's common to have way more parameters (features) than samples. We might have data from 10,000 genes but only 500 mice [18:04].
- Least Squares **cannot be solved** in this situation. There are infinite possible solutions (just like infinite lines can go through one data point) [18:43].
- **The Ridge Solution:**
  Ridge Regression *can* find a solution even when there are more parameters than data points [18:22].
  By adding the penalty $\lambda \times \sum(\beta)^2$, we introduce a new constraint that makes the problem solvable. The penalty (combined with cross-validation) favors smaller parameter values and allows the model to find a single, optimal, and stable solution where least squares would fail [18:55].
  http://googleusercontent.com/youtube_content/8